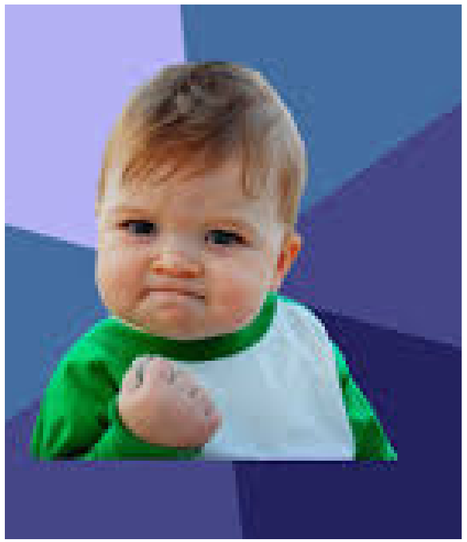
# COMPUTER NETWORKS

CSE 321

We are connected to the Wi-Fi

Now, let’s browse the Internet



# Motivation

You want to take service from a certain server (say, Google server). How to know the address of the server?

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## Solution 1

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Memorize important IPs

Very hard to remember 12 digit numbers

Version 6 has 32 hexadecimal digits

## Motivation

You want to take service from a certain server (say, Google server). How to know the address of the server?

## Solution 2

Map IPs to something easy to remember

# DNS

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Domain names are provided by accredited domain name registrars, which are companies authorized by ICANN (Internet Corporation for Assigned Names and Numbers) to sell and manage domain registrations to the public.

# Who Will Map?

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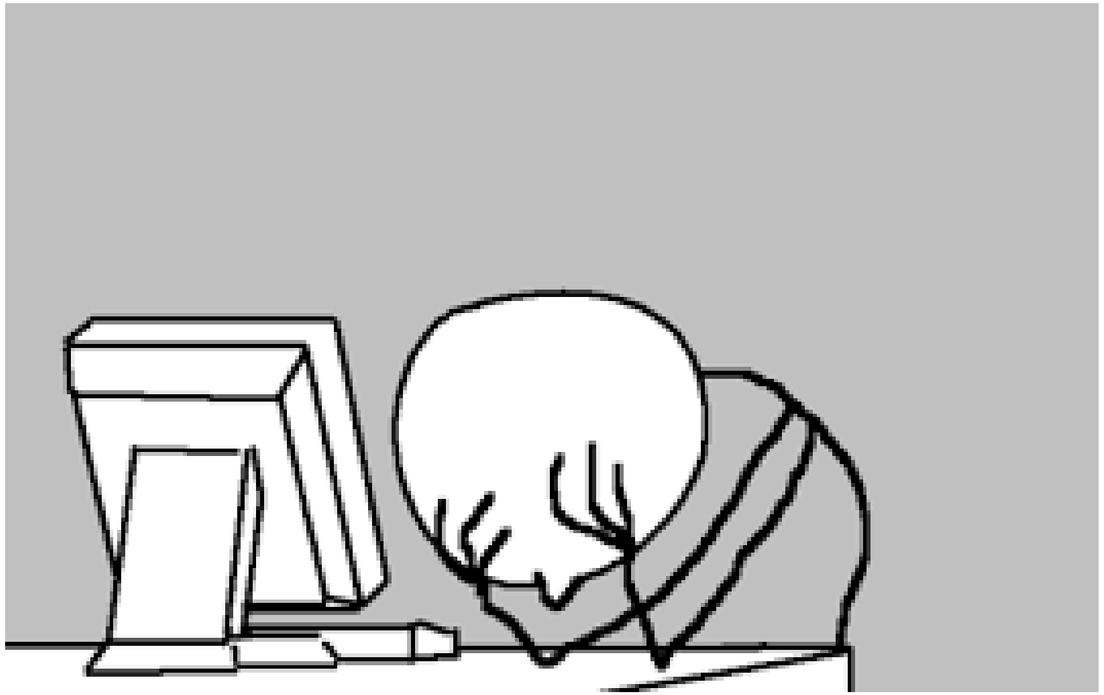
## Your PC?

# Who Will Map?

## Your PC?

## Needs HUGE storage space



# Who Will Map?

Some dedicated server

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Some dedicated server

Can be a solution.

# Who Will Map?

Some dedicated server

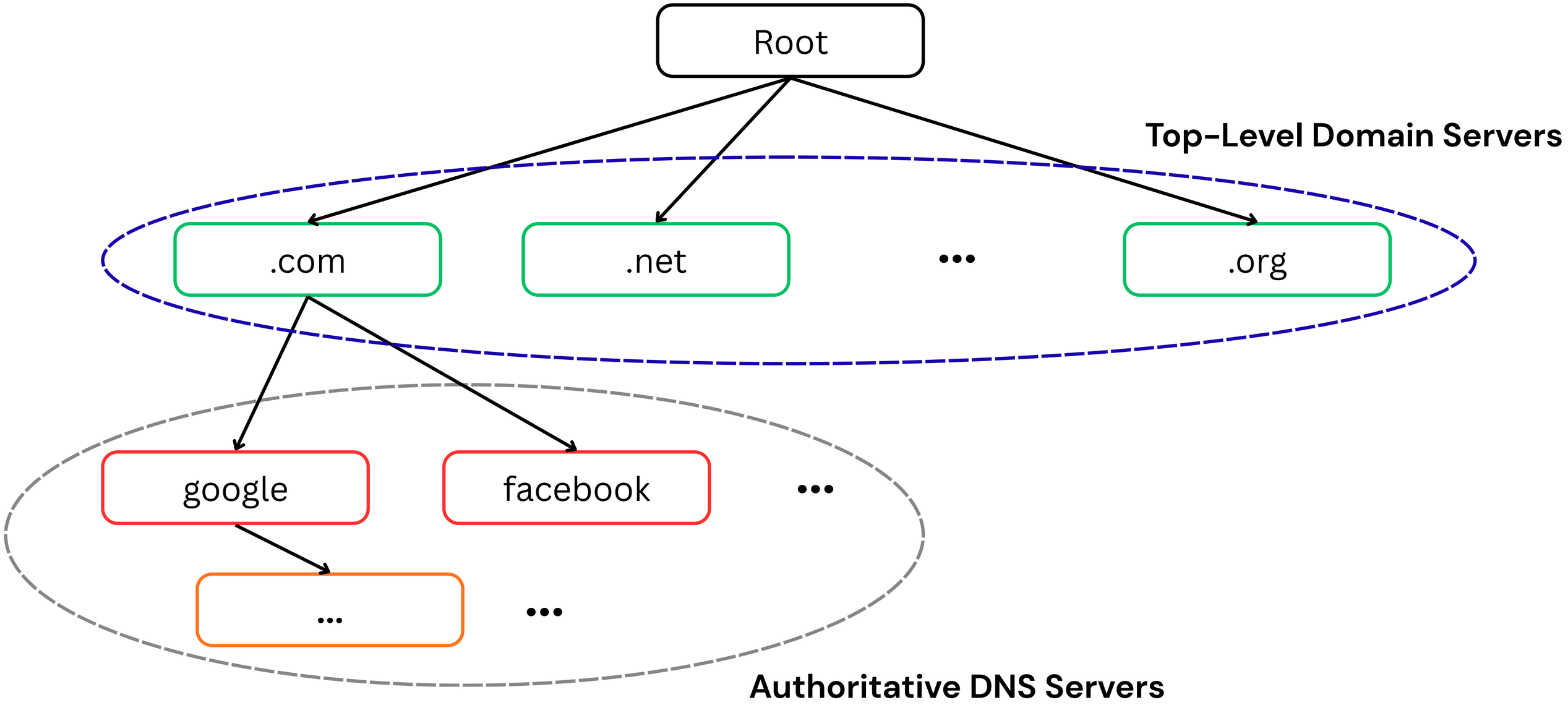
Can be a solution. But,

- Single point of failure
- Huge traffic

# Solution?

# Solution?

## Hierarchical Structure



## Solution?

## Hierarchical Structure

There are also Local DNS servers (like the DNS server of your ISP) outside the hierarchy, that cache important and frequently (or recently) used translations.

# Solution?

## Multiple Root Servers

- Fault tolerance and high availability
- Load distribution and scalability
- Reduced latency via geographic distribution

# DNS Protocol Messages

## DNS Query and DNS Reply

## Improvement

**Your PC (OS and/or Browser) also caches recent DNS translations to avoid frequent look-ups.**

## Example

Suppose you want to check your mail on mail.google.com

And, the translation is not available in your cache

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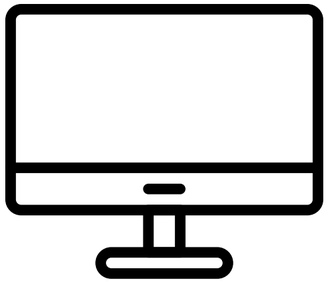
And, the translation is not available in your cache

Then we need to do DNS query. This can be done in 2 ways

# Example (Iterated Query)



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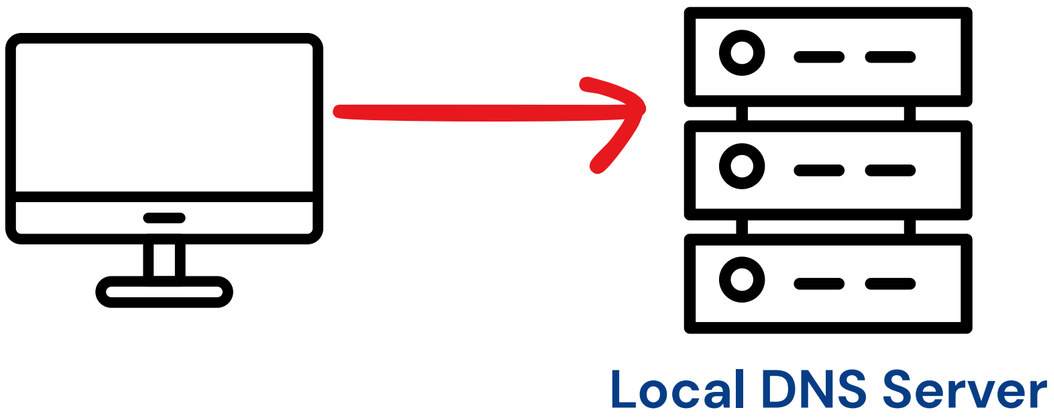

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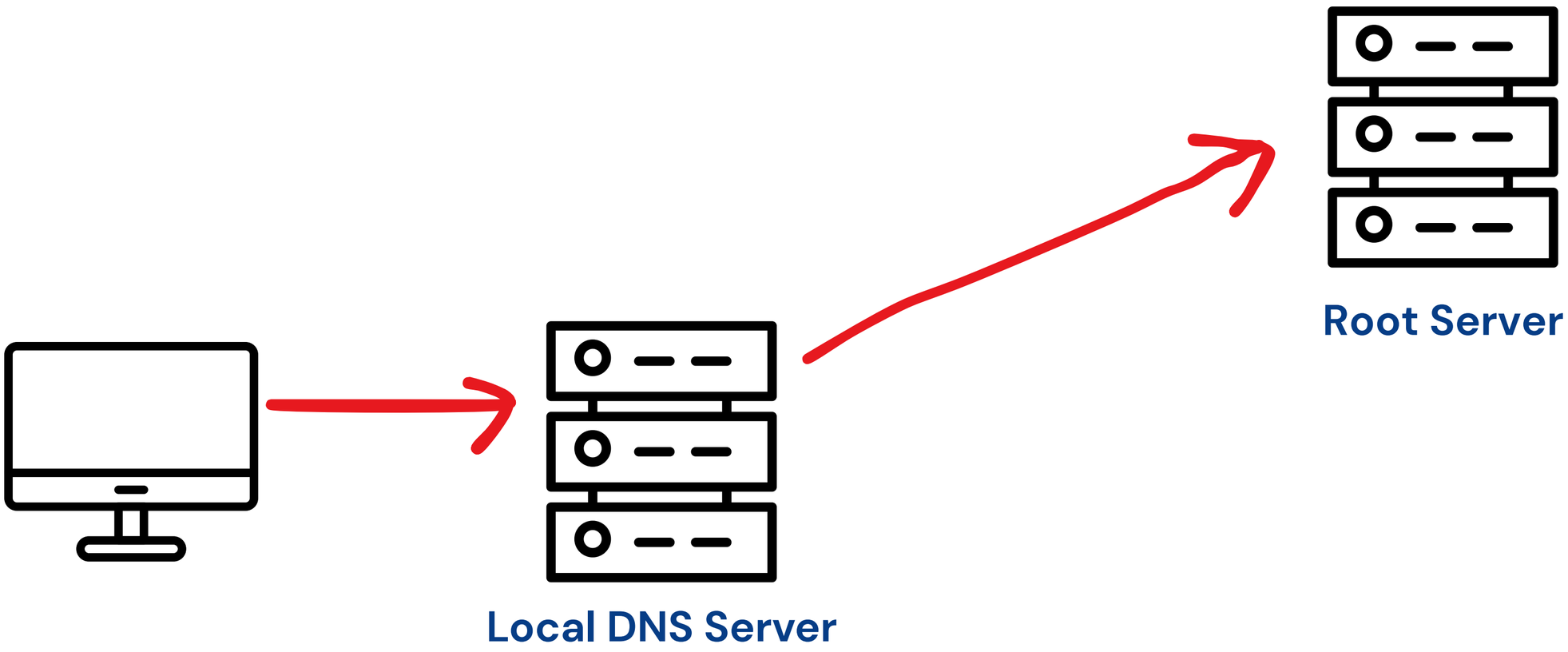

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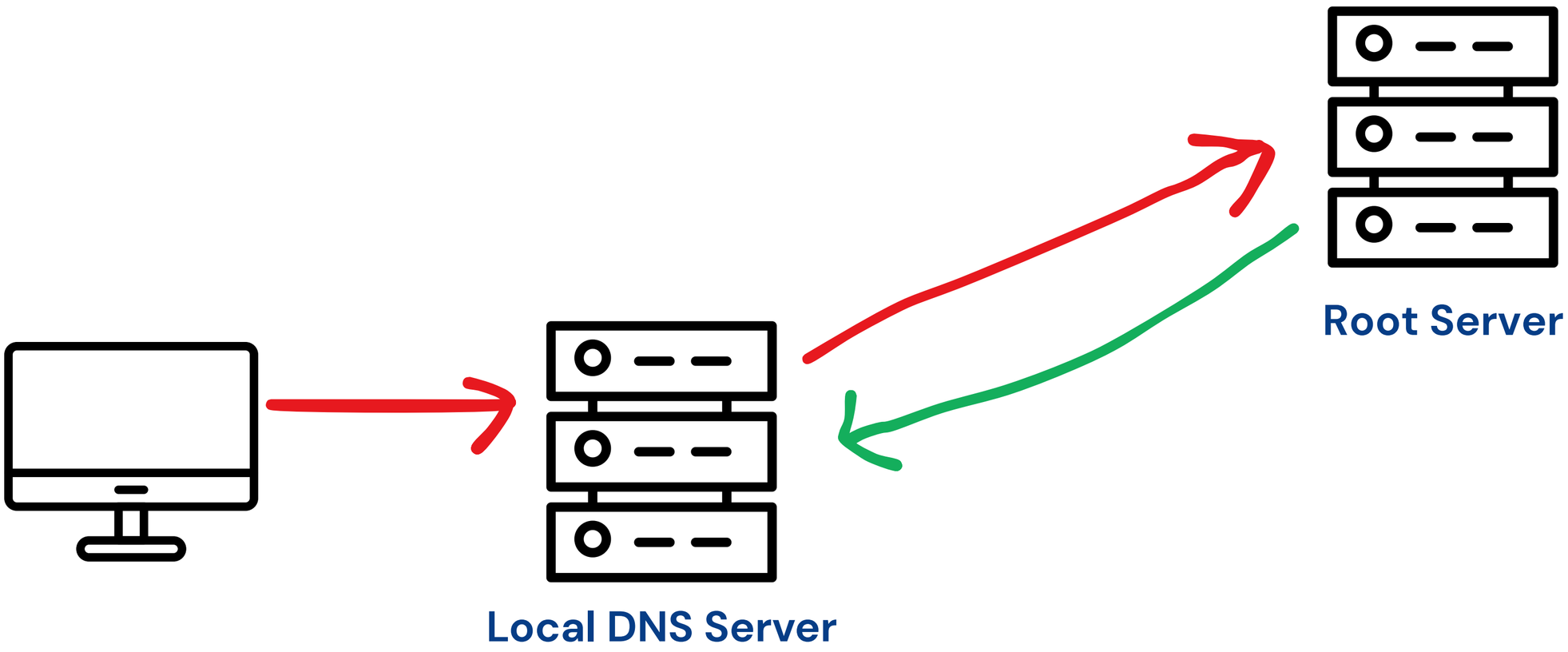

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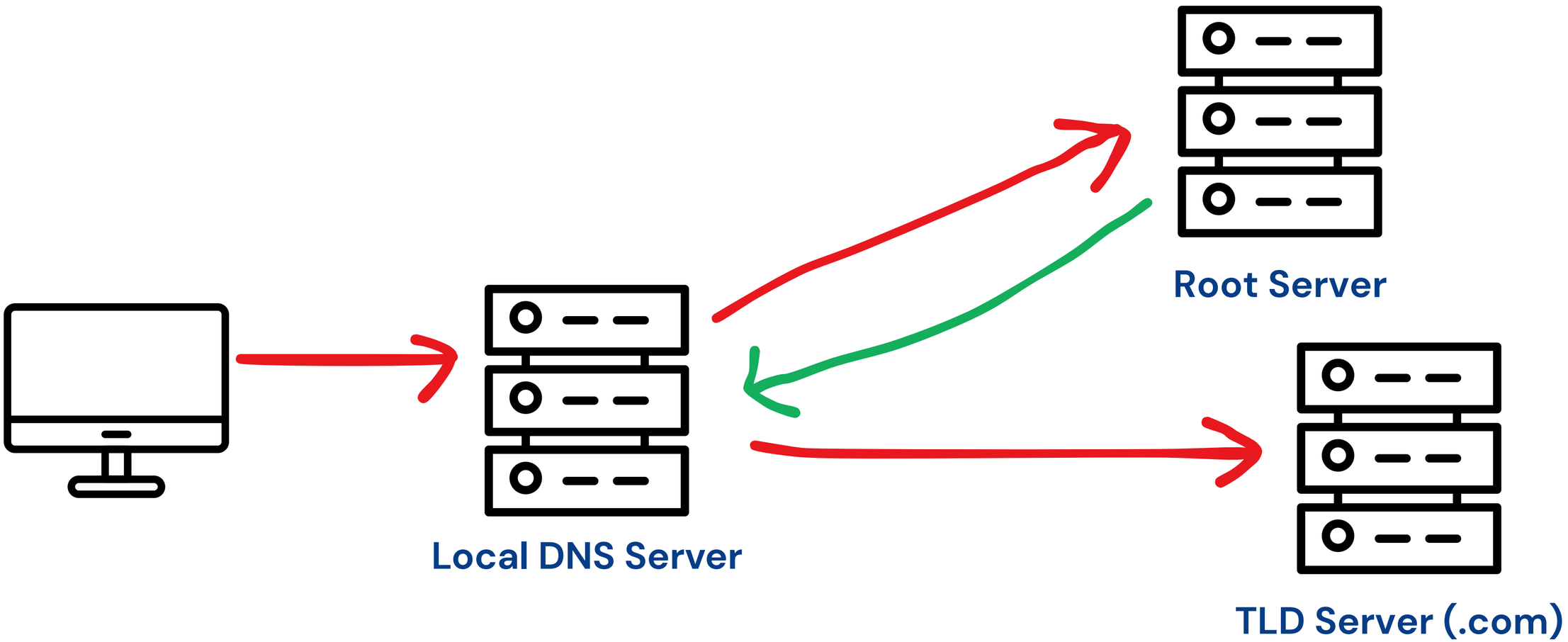

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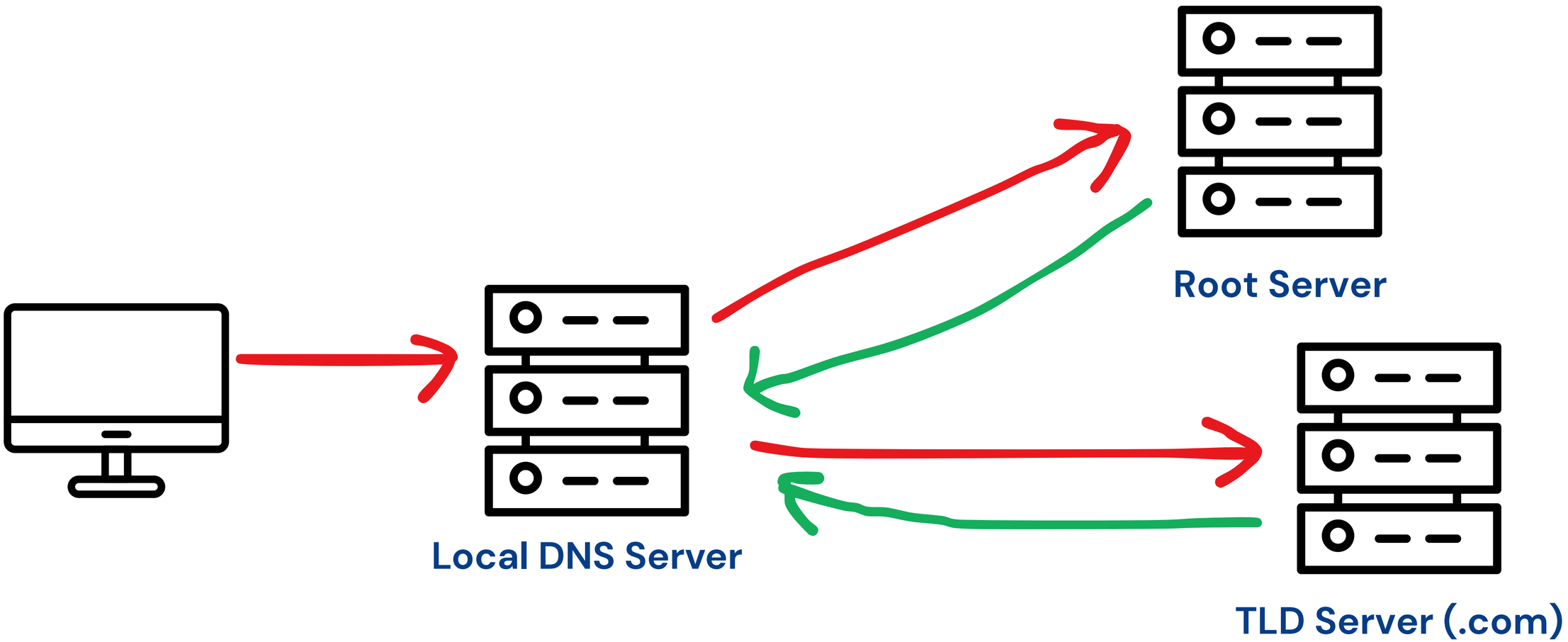

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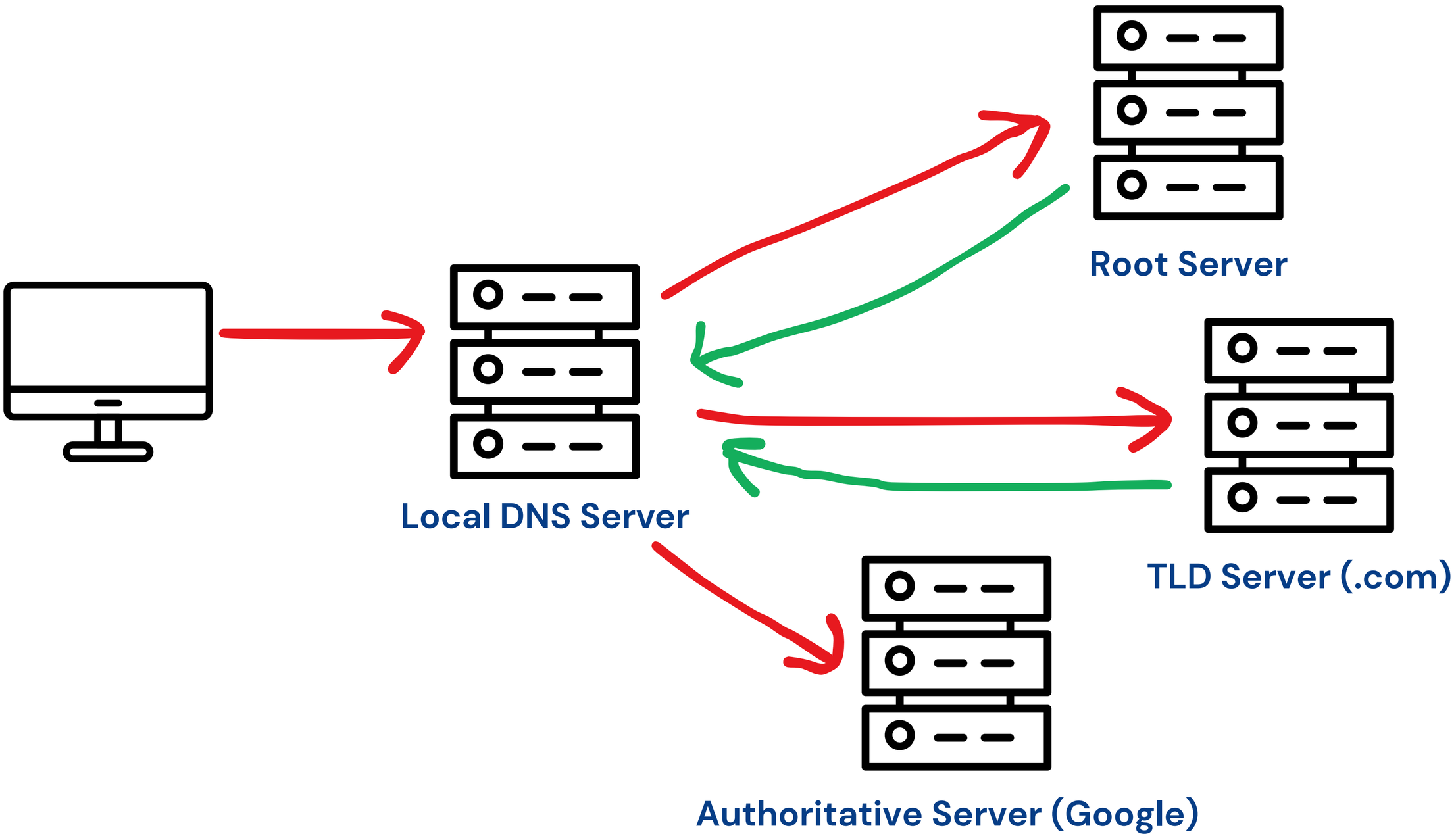

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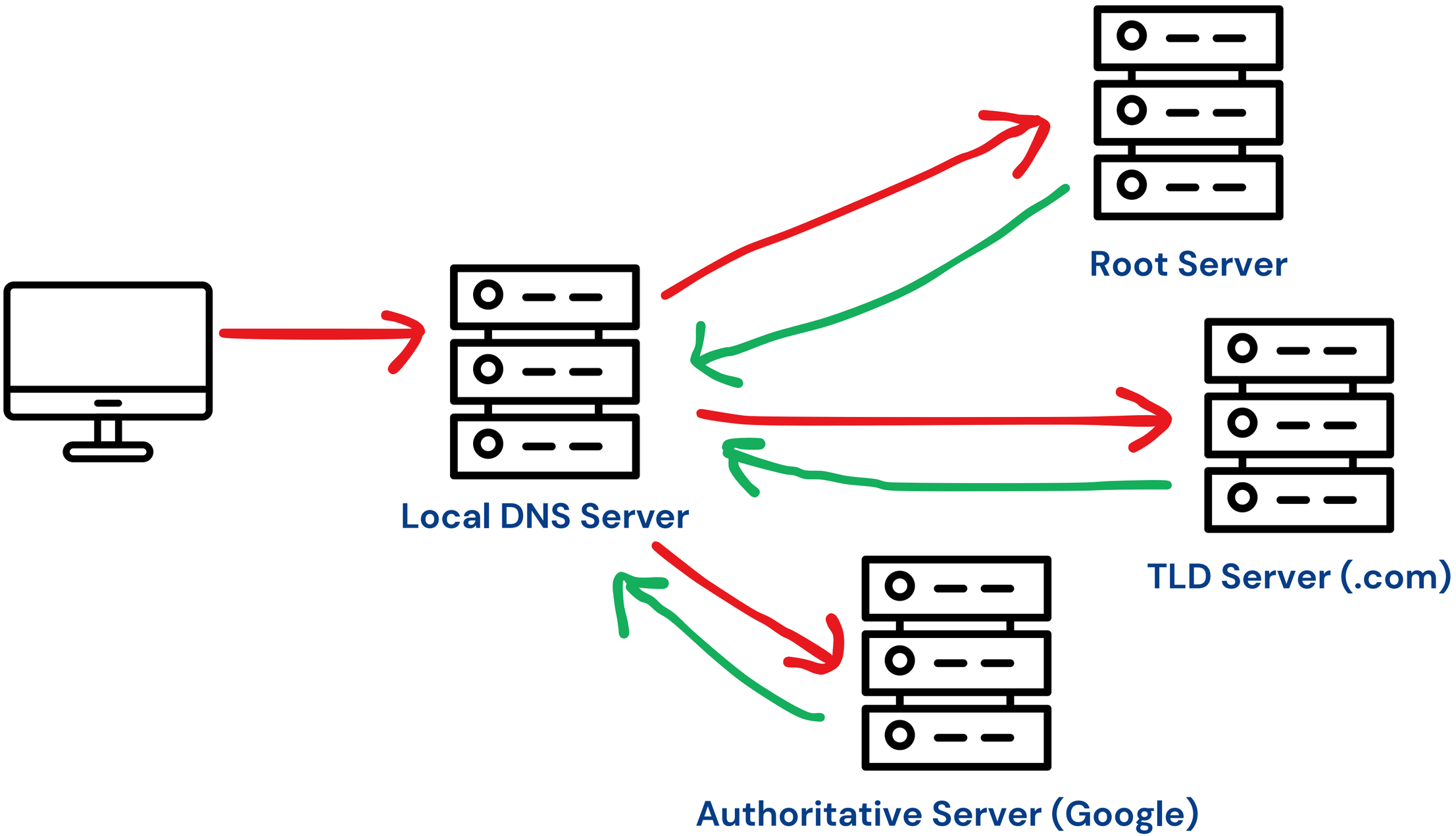

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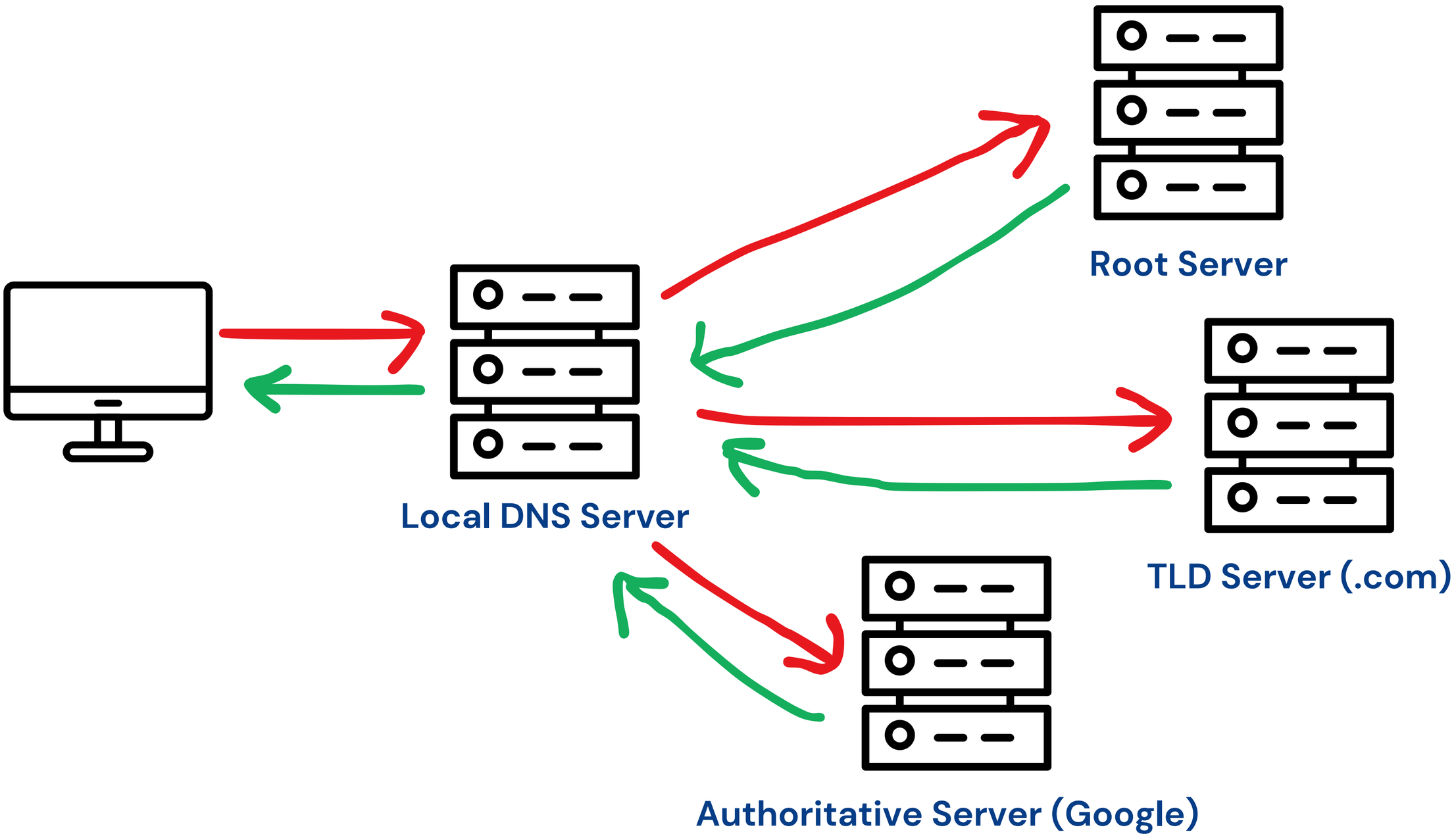

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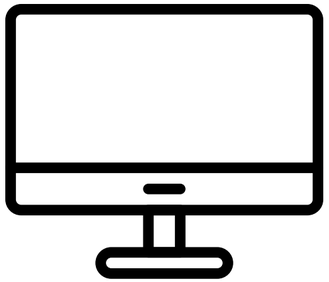

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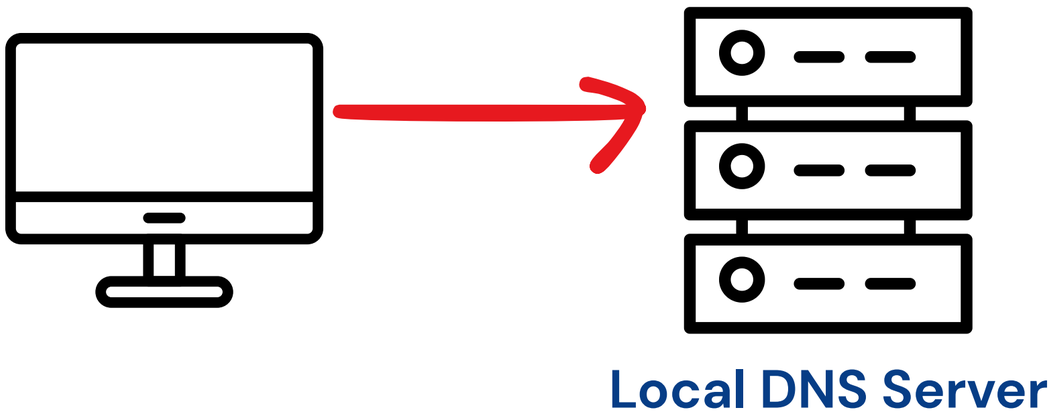

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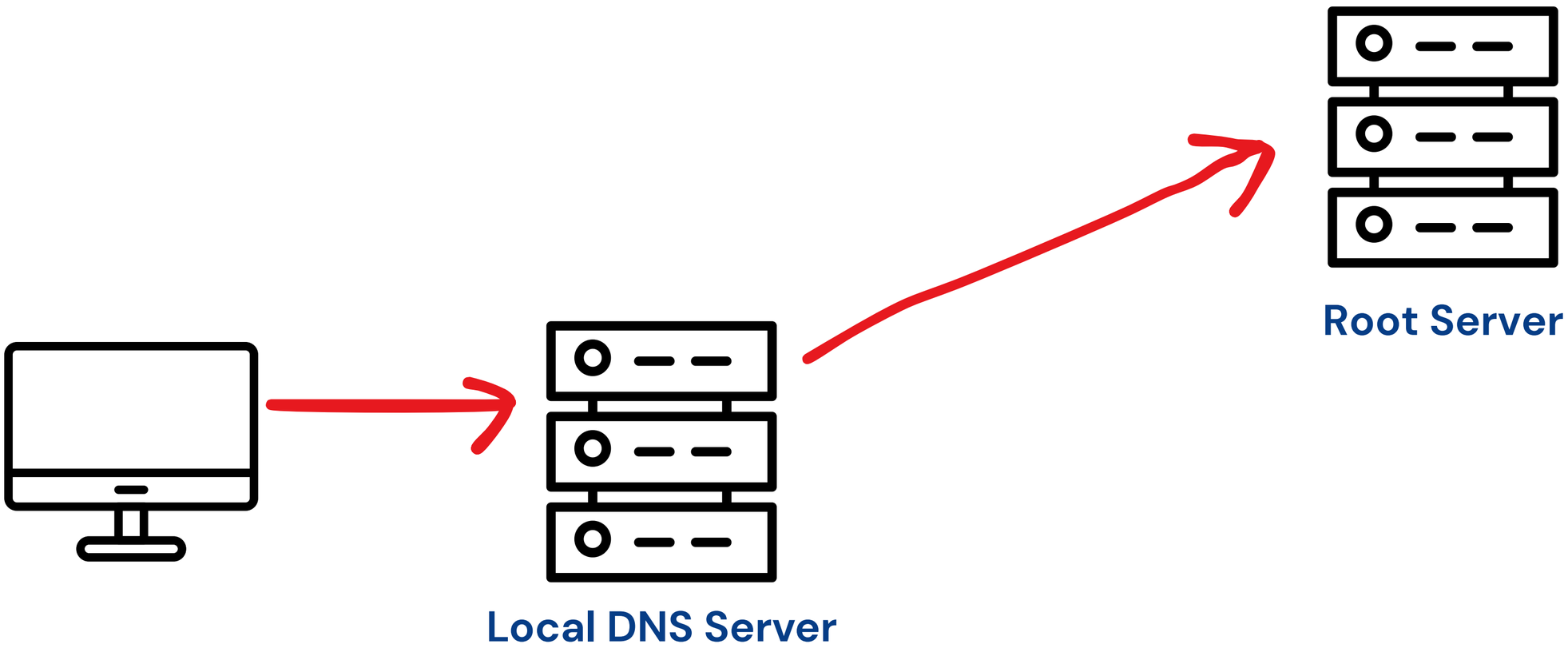

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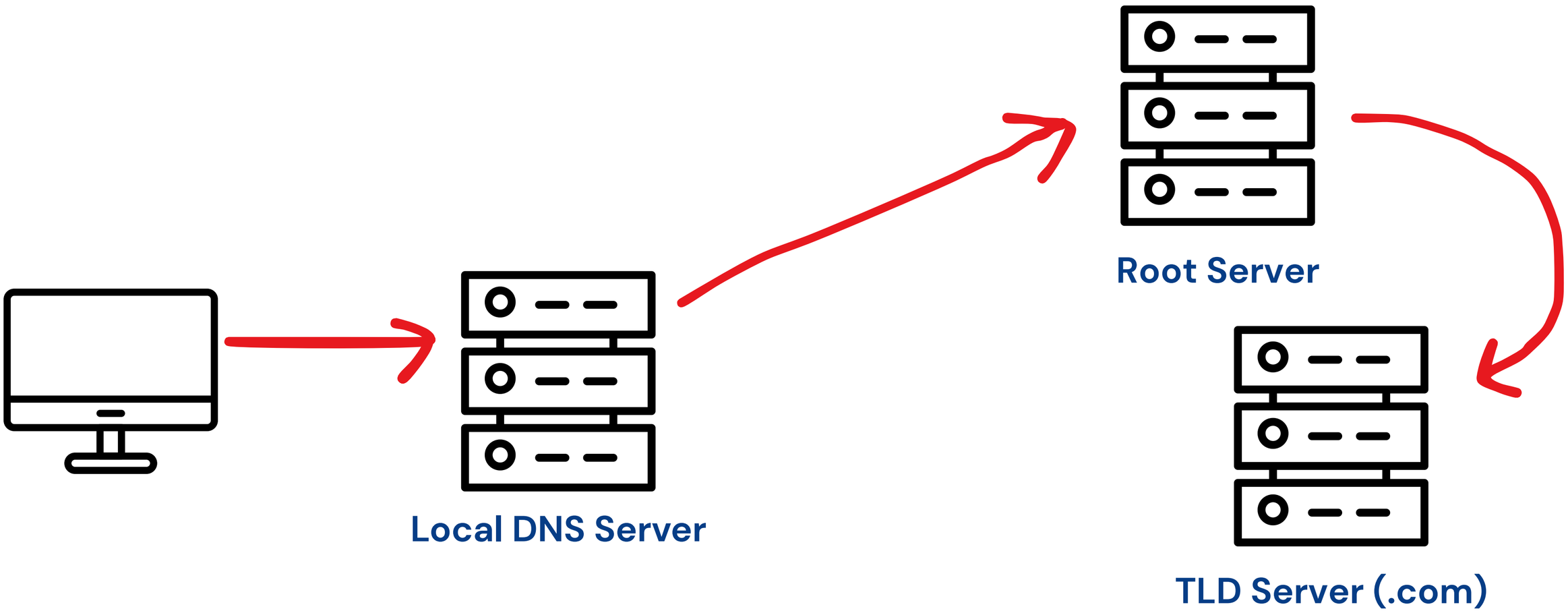

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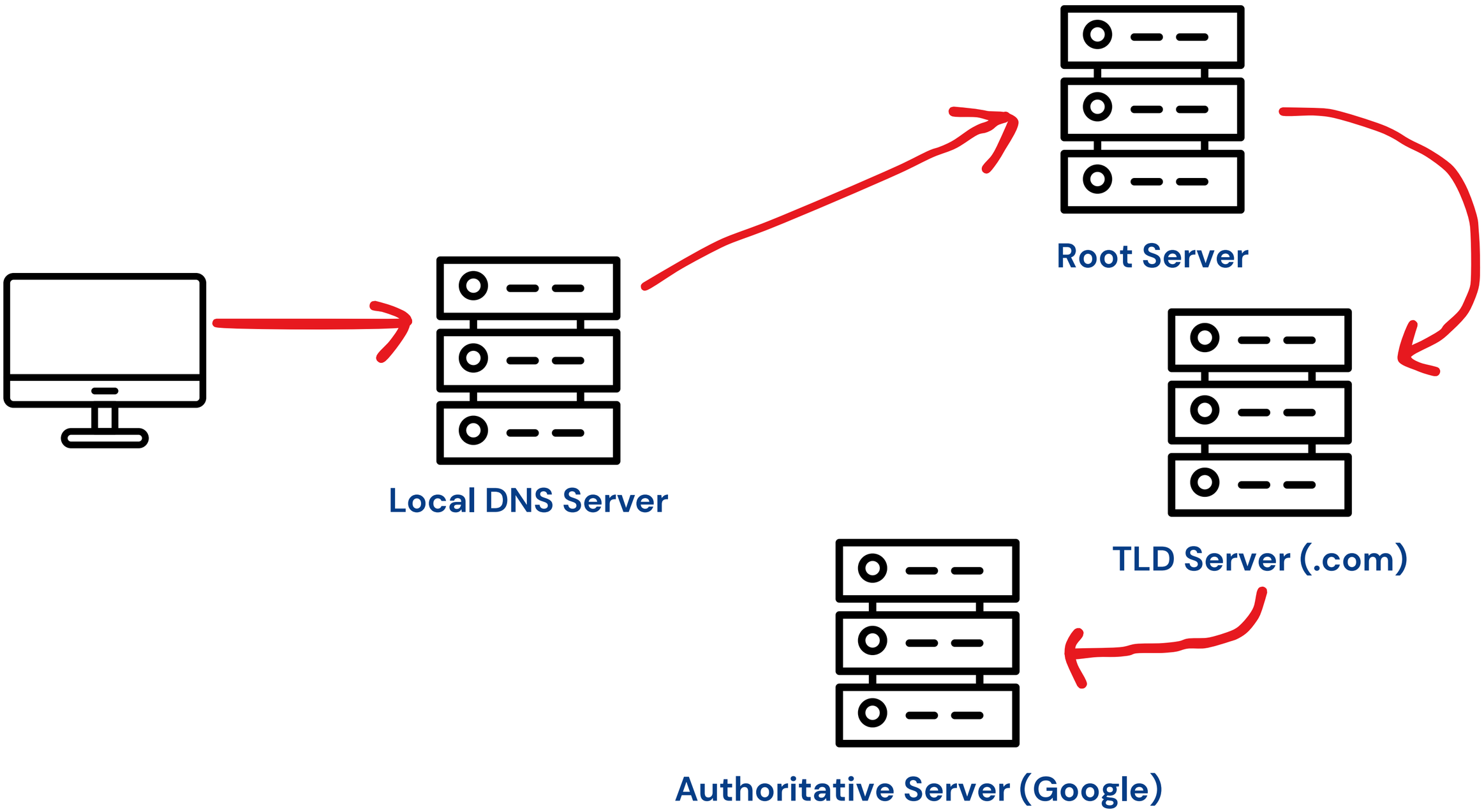

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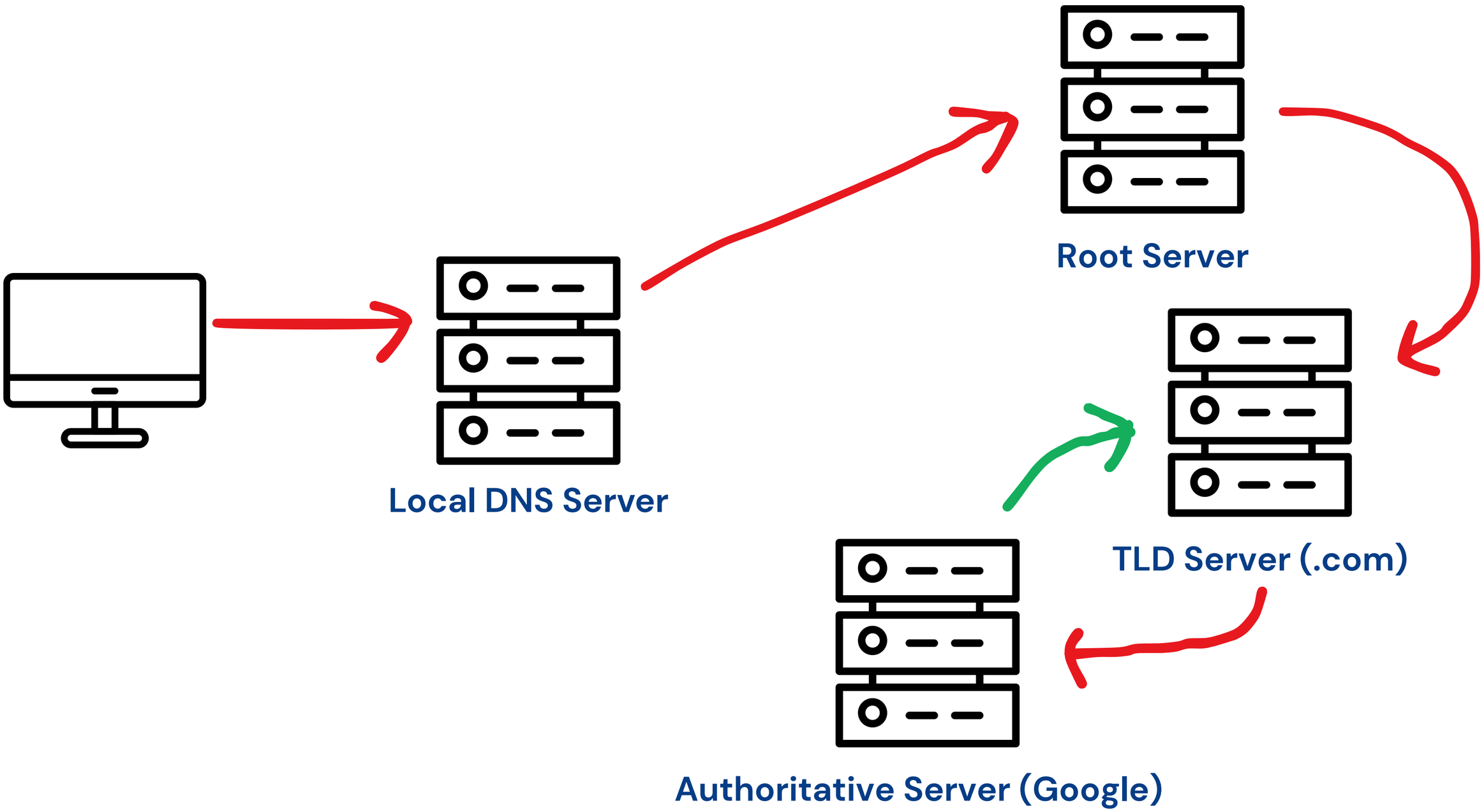

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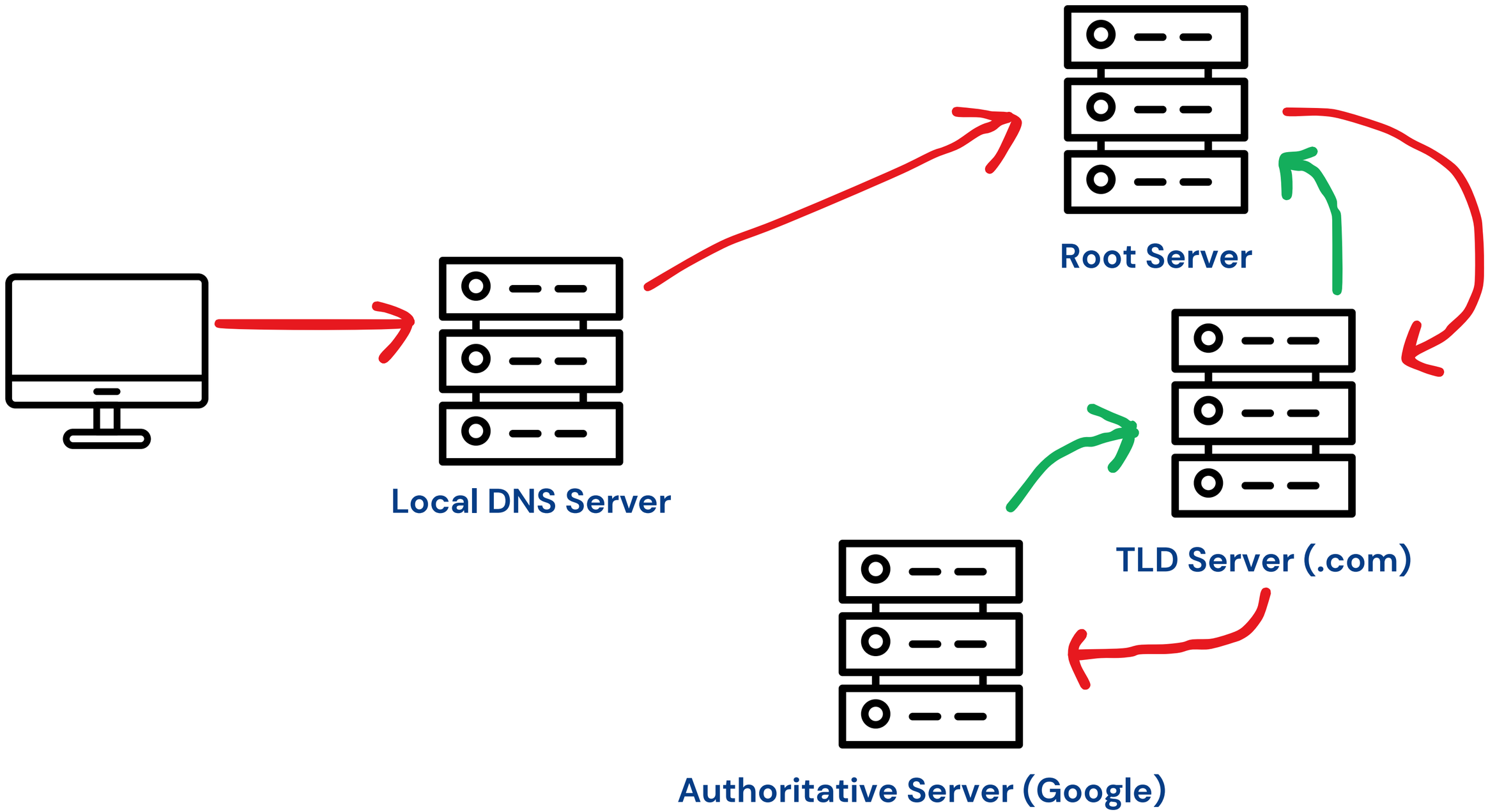

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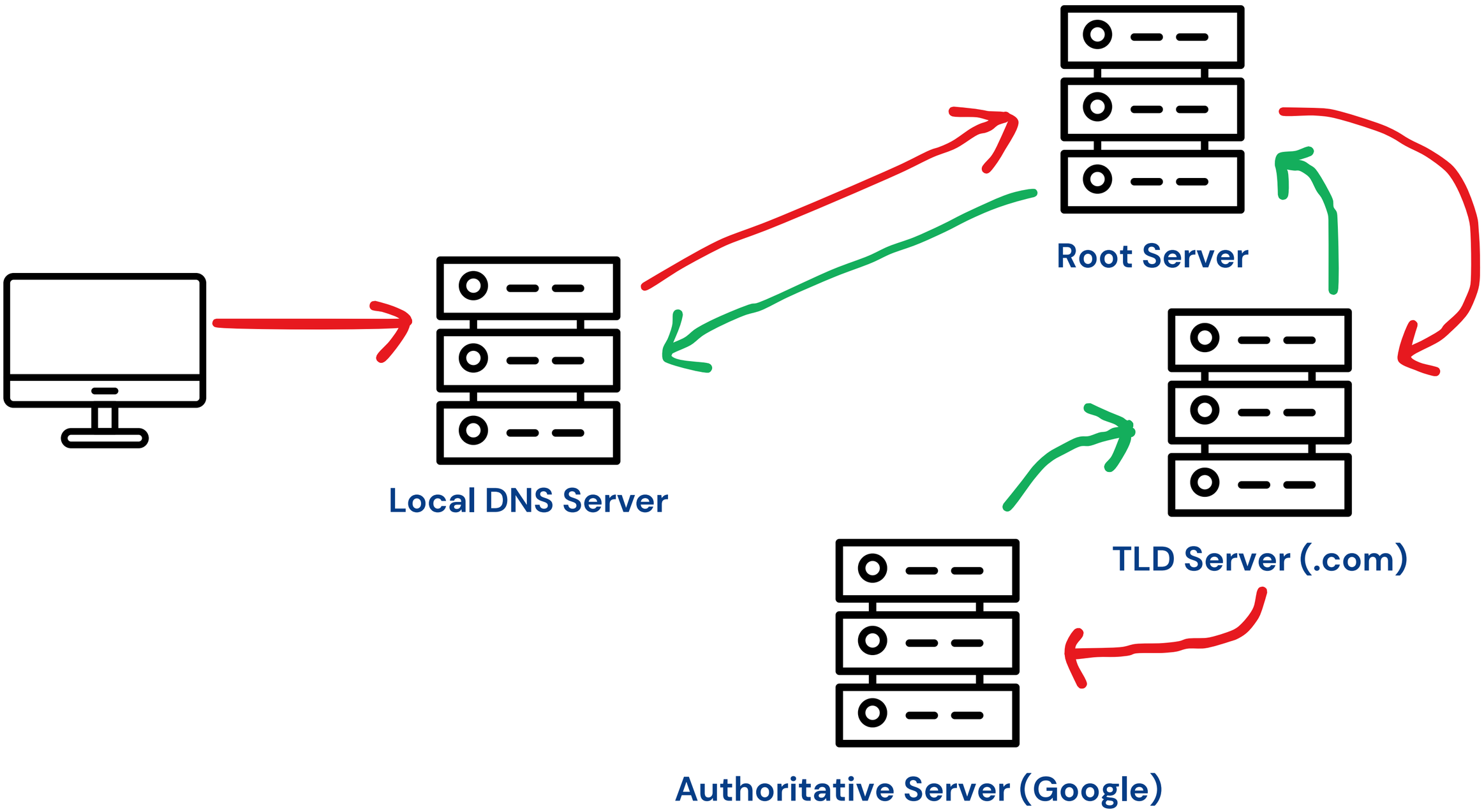

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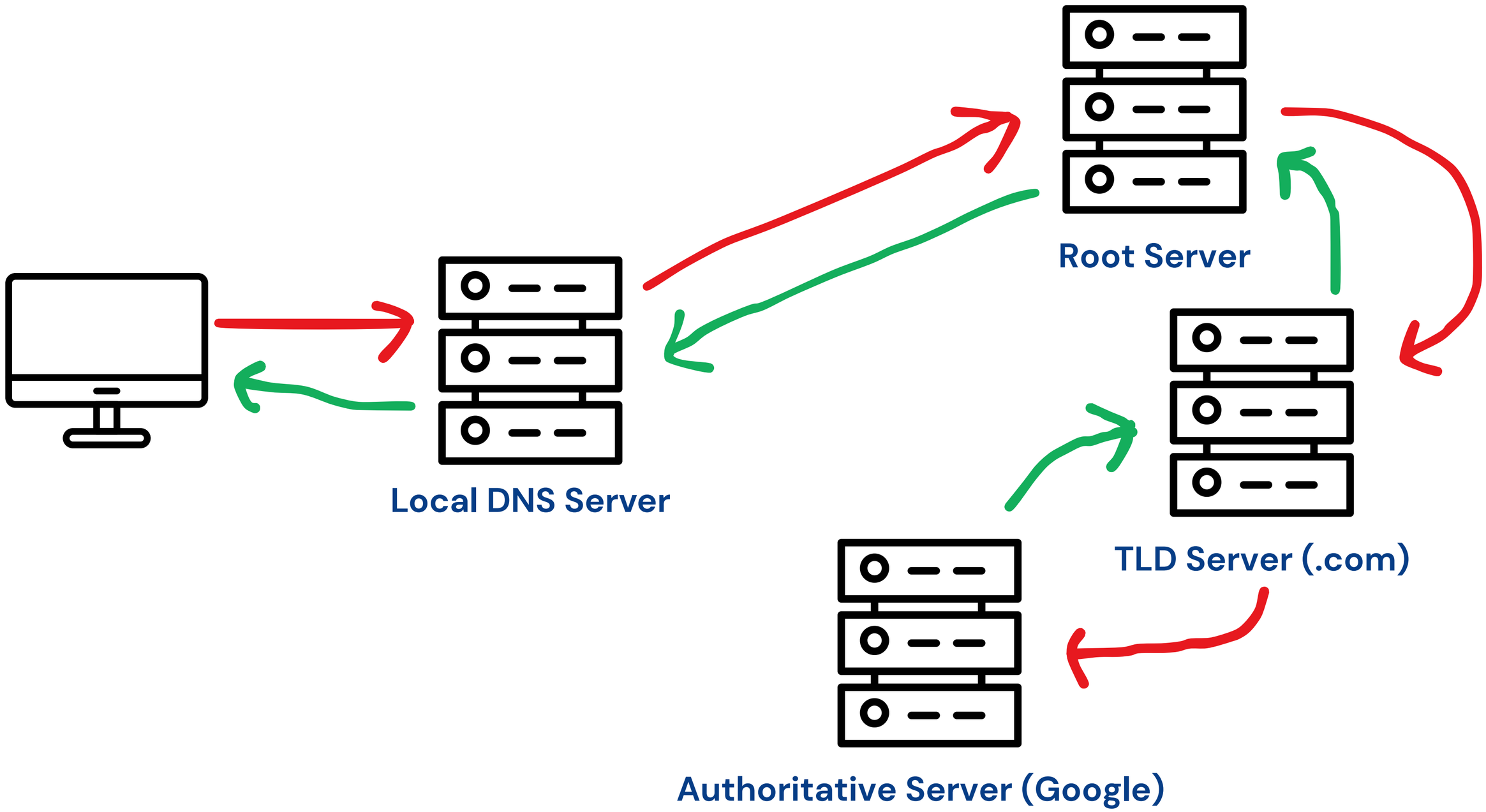

# Iterated vs Recursive

| Iterated | Recursive |
| --- | --- |
| More work for client | Easy for client |
| Lower load on upstream servers | Heavy load on upstream servers |

# DNS Records

Format → (name, value, type, ttl)

Duration for caching

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## Format → (name, value, type, ttl)

| Name | Value | Type | TTL |
|---|---|---|---|
| google.com | 129.0.2.3 | A | - |

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| google.com | 129.0.2.3 | A | - |
| gagle.com | google.com | CNAME | - |

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|---|---|---|---|
| google.com | 129.0.2.3 | A | - |
| gagle.com | google.com | CNAME | - |
| foo.com | ns1.foo.com | NS | - |

**And more...**

THE #1 DevOps EXCUSE
FOR LEGITIMATELY SLACKING OFF:
"Waiting for DNS cache"
HEY! GET BACK TO WORK!
DNS!
OH. CARRY ON.
made with mematic

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